all you need to do is figure out how many oranges are packed in each. This you easily do by borrowing your friend's box and filling it with oranges, allowing you to finish your huge counting task with almost no effort. In essence, by cleverly reorganizing the calculation, you were able to make it substantially easier to accomplish.

The situation with numerous calculations in string theory is similar. From the perspective of one Calabi-Yau space, a calculation might involve an enormous number of difficult mathematical steps. By translating the calculation to its mirror, though, the calculation is reorganized in a far more efficient manner, allowing it to be completed with relative ease. This point was made by Plesser and me, and was impressively put into practice in subsequent work by Candelas with his collaborators Xenia de la Ossa and Linda Parkes, from the University of Texas, and Paul Green, from the University of Maryland. They showed that calculations of almost unimaginable difficulty could be accomplished by using the mirror perspective, with a few pages of algebra and a desktop computer.

This was an especially exciting development for mathematicians, because some of these calculations were precisely the ones they had been stuck on for many years. String theory—or so the physicists claimed—had beaten them to the solution.

Now you should bear in mind that there is a good deal of healthy and generally good-natured competition between mathematicians and physicists. And as it turns out, two Norwegian mathematicians—Geir Ellingsrud and Stein Arilde Strømme—happened to be working on one of numerous calculations that Candelas and his collaborators had successfully conquered with mirror symmetry. Roughly speaking, it amounted to calculating the number of spheres that could be "packed" inside a particular Calabi-Yau space, somewhat like our analogy of counting oranges in a large bin. At a meeting of physicists and mathematicians in Berkeley in 1991, Candelas announced the result reached by his group using string theory and mirror symmetry: 317,206,375. Ellingsrud and Strømme announced the result of their very difficult mathematical calculation: 2,682,549,425. For days, mathematicians and physicists debated: Who was right? The question turned into a real litmus test of the quantitative reliability of string theory. A number of people even commented—somewhat in jest—that